Quasi-static viscoelastic and elasto-plastic solids on Jexpresso's spectral elements

The thick-walled cylinder and the perforated plate of
Artioli, Beirão da Veiga, Lovadina & Sacco (2017), §§4.2–4.3

Jexpresso, branch `sm/solid`, commit `2ab96bf`

Abstract

This note documents the quasi-static solid-mechanics path of Jexpresso (`:lquasistatic_solid`) and the benchmark it was built for: a plane-strain thick-walled cylinder under a step internal pressure, made of a generalized Maxwell viscoelastic material (Artioli et al. 2017, Secs. 3.1 and 4.2). It states the governing equations, the constitutive laws (the viscoelastic one of the benchmark and the J_2 plasticity that uses the same quadrature-point history), every step of the spectral-element discretisation, and the initialisation, and links each to the line of the Jexpresso source where it is done. Section 7 adds the paper's second inelastic benchmark, a perforated elastic–perfectly plastic plate (§4.3), and the script that redraws the paper's figures from Jexpresso's VTK output. Every link is a permalink to commit `2ab96bf`.

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1 The problem

A long cylinder of inner radius $a = 2$ and outer radius $b = 4$ is loaded by an internal pressure $p = 10$ switched on at $t = 0$ and held to $t = 20$; the outer surface is free. The cylinder is long, so the state

is *plane strain* ($u_z = 0$, $\varepsilon_{zz} = 0$), and by symmetry one quarter of the cross-section is modelled, with rollers on the two straight edges (Fig. 1). The control points are $A = (a, 0)$ on the bore and $B = (b, 0)$ on the outer surface, where $u_r = u_x$.

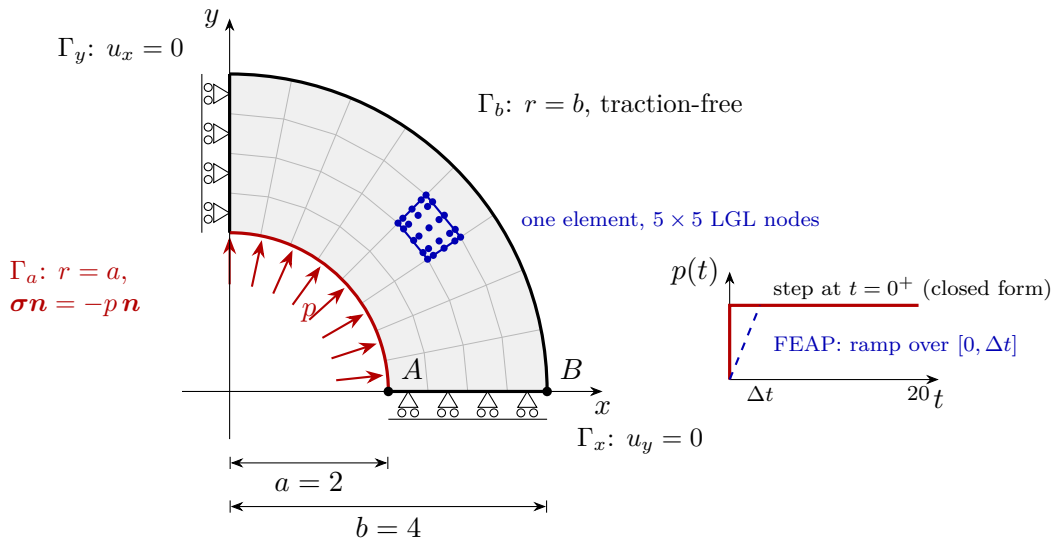


Figure 1: The quarter cylinder (plane strain) and its boundary conditions, drawn over the spectral-element grid of the case: 4×8 quadrilaterals, both arcs curved onto their circles; one element shows its 5×5 LGL nodes ($N = 4$). Inset: the pressure history — the Heaviside step of the closed form, and the ramp over the first increment that the paper's FEAP runs amount to (§5).

quantity	value	note
Young's modulus, Poisson's ratio	$E = 1000$, $\nu = 0.3$	instantaneous
shear, bulk modulus	$G = E/2(1+\nu) = 384.6$, $K = E/3(1-2\nu) = 833.3$	
Prony series	$G(t) = G\,(\mu_0 + \mu_1 e^{-t/\lambda})$, $\lambda = 1$	λ: relaxation time
set ve1 / ve2	$(\mu_0, \mu_1) = (0.01, 0.99)$ / $(0.3, 0.7)$	`JEXPRESSO_VE_SET`
geometry, load	$a = 2$, $b = 4$, $p = 10$, $t \in [0, 20]$	
discretisation	$N = 4$, 4×8 elements, $\Delta t = 1$	1122 DOFs

Table 1: Data of the benchmark. Note: the paper's λ is a *relaxation time*; it has nothing to do with the Lamé constant, which is why the code calls it τ.

Code: the case: `problems/Elasticity/viscoelastic_cylinder2d`; deck `user_inputs`; material, geometry, load `cylinder_properties`; mesh `structured_quarter_annulus`.

2 Governing equations

Small strains, no inertia (the loading is slow compared with the elastic wave transit, so the problem is *quasi-static*), no body force. For every $t \in [0, T]$ find the displacement $\boldsymbol{u}(\boldsymbol{x}, t)$ on the quarter

cross-section Ω such that

$$\nabla\cdot\boldsymbol{\sigma}+\boldsymbol{b}=\boldsymbol{0} \qquad \text{in } \Omega, \tag{1}$$

$$\boldsymbol{\varepsilon}(\boldsymbol{u})=\operatorname{sym}\nabla\boldsymbol{u}=\tfrac{1}{2}(\nabla\boldsymbol{u}+\nabla\boldsymbol{u}^{\mathsf{T}}),\quad \varepsilon_{zz}=0 \qquad \text{(plane strain)}, \tag{2}$$

$$\boldsymbol{\sigma}\boldsymbol{n}=-p(t)\,\boldsymbol{n} \qquad \text{on } \Gamma_a\ (r=a), \tag{3}$$

$$\boldsymbol{\sigma}\boldsymbol{n}=\boldsymbol{0} \qquad \text{on } \Gamma_b\ (r=b), \tag{4}$$

$$u_y=0,\quad (\boldsymbol{\sigma}\boldsymbol{n})_x=0 \qquad \text{on } \Gamma_x\ (y=0), \tag{5}$$

$$u_x=0,\quad (\boldsymbol{\sigma}\boldsymbol{n})_y=0 \qquad \text{on } \Gamma_y\ (x=0), \tag{6}$$

with $\boldsymbol{n}$ the outward unit normal and $\boldsymbol{\sigma}$ given by the constitutive law of §3, which carries the memory of the material. Time enters *only* through that memory and through $p(t)$.

Weak form. With $V=\{\boldsymbol{v}\in H^1(\Omega)^2 : v_y=0 \text{ on } \Gamma_x,\ v_x=0 \text{ on } \Gamma_y\}$, find $\boldsymbol{u}(t)\in V$ such that

$$\int_\Omega \boldsymbol{\varepsilon}(\boldsymbol{v}):\boldsymbol{\sigma}\,\mathrm{d}\Omega=\int_{\Gamma_a}\boldsymbol{v}\cdot(-p\,\boldsymbol{n})\,\mathrm{d}\Gamma+\int_\Omega \boldsymbol{v}\cdot\boldsymbol{b}\,\mathrm{d}\Omega \qquad \forall\,\boldsymbol{v}\in V. \tag{7}$$

Only the in-plane components of $\boldsymbol{\sigma}$ appear ($\varepsilon_{zz}(\boldsymbol{v})=0$), but the material point is three-dimensional and σ_{zz} is part of the solution.

Code: boundary conditions `user_solid_bc`; body force `user_solid_body_force`; the solver and its documentation `quasistatic_solid.jl`; dispatch from the driver `drivers.jl`.

3 Constitutive laws

3.1 Generalized Maxwell viscoelasticity (the benchmark)

Volumetric–deviatoric split, $\boldsymbol{\varepsilon}=\boldsymbol{e}+\frac{1}{3}\operatorname{tr}(\boldsymbol{\varepsilon})\mathbf{1}$, elastic in volume and generalized Maxwell in the deviator (Artioli et al. eqs. 3.2–3.7):

$$\boldsymbol{\sigma}=K\,\operatorname{tr}(\boldsymbol{\varepsilon})\,\mathbf{1}+\boldsymbol{s},\qquad \boldsymbol{s}(t)=\int_{-\infty}^{t}2G(t-s)\,\dot{\boldsymbol{e}}(s)\,\mathrm{d}s,\qquad G(t)=G\Big(\mu_0+\sum_{m=1}^{M}\mu_m e^{-t/\lambda_m}\Big), \tag{8}$$

with $\mu_0+\sum_m\mu_m=1$. Each Prony term is a spring–dashpot branch whose spring strain $\boldsymbol{q}_m$ (a deviatoric tensor) obeys

$$\dot{\boldsymbol{q}}_m+\frac{1}{\lambda_m}\boldsymbol{q}_m=\dot{\boldsymbol{e}},\qquad \boldsymbol{s}=2G\Big(\mu_0\,\boldsymbol{e}+\sum_m\mu_m\boldsymbol{q}_m\Big). \tag{9}$$

In plane strain $e_{zz}=-\frac{1}{3}\operatorname{tr}\boldsymbol{\varepsilon}\neq 0$, so $\boldsymbol{q}_m$ and $\boldsymbol{s}$ have four non-zero components (xx,yy,zz,xy) and $\sigma_{zz}=K\operatorname{tr}\boldsymbol{\varepsilon}+s_{zz}$ creeps.

Time integration: the exponential recursion. Integrating (9) exactly over $[t_n,t_{n+1}]$ for a strain that is linear in time (Zienkiewicz, Taylor & Fox 2014; Simo & Hughes 1998, §10.3) gives

$$\boldsymbol{q}_m^{n+1}=d_m\,\boldsymbol{q}_m^{n}+\gamma_m(\boldsymbol{e}^{n+1}-\boldsymbol{e}^{n}),\qquad d_m=e^{-\Delta t/\lambda_m},\quad \gamma_m=\frac{\lambda_m}{\Delta t}(1-e^{-\Delta t/\lambda_m}), \tag{10}$$

second-order accurate; it is the update FEAP — and so the paper — uses. Simo & Hughes' Box 10.1 variant $\gamma_m = e^{-\Delta t/2\lambda_m}$ is also second order; backward Euler $d_m = \gamma_m = (1+\Delta t/\lambda_m)^{-1}$ is first order. Because (10) is linear in $\boldsymbol{e}^{n+1}$,

$$\boldsymbol{\sigma}^{n+1} = K\,\mathrm{tr}(\boldsymbol{\varepsilon}^{n+1})\mathbf{1} + 2\tilde{G}\,\boldsymbol{e}^{n+1} + \boldsymbol{\sigma}^*, \qquad \tilde{G} = G\Big(\mu_0 + \sum_m \mu_m\gamma_m\Big), \quad \boldsymbol{\sigma}^* = 2G\sum_m \mu_m\big(d_m\boldsymbol{q}_m^n - \gamma_m\boldsymbol{e}^n\big)\,: \tag{11}$$

each step is a linear elastic problem with the *algorithmic* shear modulus $\tilde{G}$ and a known initial stress $\boldsymbol{\sigma}^*$. For $\Delta t = 0$, $d_m = \gamma_m = 1$: the instantaneous (glassy) response.

Code: coefficients (d_m, γ_m) `maxwell_step_coefficients`; the material point `solid_constitutive_update!`; the case's material `user_solid_material`.

Closed-form solution. By the correspondence principle (load p/s, $G \to s\hat{G}(s)$, K unchanged), for the Heaviside pressure

$$u_r(r,t) = P\Big[r\,C(t) + \frac{b^2}{2r}J(t)\Big], \qquad P = \frac{p\,a^2}{b^2-a^2}, \tag{12}$$

$$J(t) = \frac{1}{G}\Big[\frac{1}{\mu_0} - \Big(\frac{1}{\mu_0} - 1\Big)e^{-\mu_0 t/\lambda}\Big] \quad \text{(shear creep compliance)}, \tag{13}$$

$$C(t) = \frac{1}{M_\infty} - \Big(\frac{1}{M_\infty} - \frac{1}{M_0}\Big)e^{-M_\infty t/(M_0\lambda)}, \qquad M_0 = 2K + \tfrac{2}{3}G, \quad M_\infty = 2K + \tfrac{2}{3}\mu_0 G. \tag{14}$$

At $t = 0$ it is the Lamé solution $P\,\frac{1+\nu}{E}\big[(1-2\nu)r + b^2/r\big]$. $\sigma_{rr} = P(1-b^2/r^2)$ and $\sigma_{\theta\theta} = P(1+b^2/r^2)$ do not depend on the material (both surfaces carry tractions), so they are the Lamé stresses at every t; what creeps is u_r and $\sigma_{zz}(t) = 2P\big[\nu_\infty + (\nu - \nu_\infty)e^{-t/\tau_z}\big]$, with $\nu_\infty = \frac{3K-2\mu_0 G}{2(3K+\mu_0 G)}$, $\tau_z = \lambda\frac{3K+G}{3K+\mu_0 G}$.

Code: `cylinder_ur_exact`, `cylinder_szz_exact`, and the exact-in-space time-discrete reference `cylinder_time_discrete_reference`.

3.2 J_2 plasticity: the history at the quadrature points

The same path carries models whose history is *not* a linear function of the displacement, and stores it at every quadrature point. Von Mises plasticity with linear isotropic and kinematic hardening (Artioli et al. §3.2; Simo & Hughes Box 3.2):

$$\boldsymbol{s} = 2G(\boldsymbol{e} - \boldsymbol{e}^p), \quad \boldsymbol{\alpha} = H_{\mathrm{kin}}\,\boldsymbol{e}^p, \quad \boldsymbol{X} = \boldsymbol{s} - \boldsymbol{\alpha}, \quad f = \|\boldsymbol{X}\| - \sqrt{\tfrac{2}{3}}(\sigma_{y0} + H_{\mathrm{iso}}\,\bar{e}^p) \le 0, \tag{15}$$

$$\dot{\boldsymbol{e}}^p = \dot{\zeta}\,\frac{\boldsymbol{X}}{\|\boldsymbol{X}\|}, \qquad \dot{\bar{e}}^p = \sqrt{\tfrac{2}{3}}\,\dot{\zeta}, \qquad \dot{\zeta} \ge 0,\ \dot{\zeta} f = 0. \tag{16}$$

History per point: $\boldsymbol{e}^p$ (four components) and $\bar{e}^p$. Backward Euler gives the radial return: trial $\boldsymbol{X}^{\mathrm{tr}} = 2G(\boldsymbol{e}^{n+1} - \boldsymbol{e}^{p,n}) - H_{\mathrm{kin}}\boldsymbol{e}^{p,n}$; if $f^{\mathrm{tr}} > 0$, $\Delta\zeta = f^{\mathrm{tr}}/(2G + H_{\mathrm{kin}} + \frac{2}{3}H_{\mathrm{iso}})$, $\boldsymbol{n} = \boldsymbol{X}^{\mathrm{tr}}/\|\boldsymbol{X}^{\mathrm{tr}}\|$, $\boldsymbol{e}^{p,n+1} = \boldsymbol{e}^{p,n} + \Delta\zeta\boldsymbol{n}$, and the consistent tangent is

$$\mathbb{C} = K\,\mathbf{1}\otimes\mathbf{1} + 2G\theta\,\mathbb{I}_{\mathrm{dev}} - 2G\bar{\theta}\,\boldsymbol{n}\otimes\boldsymbol{n}, \quad \theta = 1 - \frac{2G\Delta\zeta}{\|\boldsymbol{X}^{\mathrm{tr}}\|}, \quad \bar{\theta} = \frac{2G}{2G + H_{\mathrm{kin}} + \frac{2}{3}H_{\mathrm{iso}}} - (1-\theta). \tag{17}$$

A shape-memory alloy (Artioli et al. §3.3) keeps the transformation strain $\boldsymbol{e}^{tr}$ where J_2 keeps $\boldsymbol{e}^p$: the storage, the Newton loop of Step 7 (§4) and the output are the same.

Code: `solid_constitutive_update!` (J_2); history layout and plottable fields `solid_history_fields`; the plasticity case `problems/Elasticity/plastic_cylinder2d`.

4 Spectral-element discretisation, step by step

Every spatial operator below is Jexpresso's own; the solver adds the material and the time stepping.

Step 1 — the grid

gmsh writes a structured, *straight-sided* grid of 4×8 quadrilaterals with the physical groups `inner`, `outer`, `xaxis`, `yaxis` and `domain`. Jexpresso reads it, places on each element the $(N+1)^2$ Legendre–Gauss–Lobatto (LGL) nodes $\boldsymbol{\xi}_{ij} = (\xi_i, \xi_j)$ of degree $N = 4$, and numbers the nodes globally through the connectivity $\mathrm{conn}(e, i, j)$ (`connijk`). gmsh leaves every high-order node of an arc on its chord; `:exact_geometry` snaps the boundary nodes onto the circles $r = a$ and $r = b$ and blends the correction into the boundary elements (Gordon–Hall), so the boundary is the isoparametric degree-N interpolant of the circle.

Code: `make_meshes.py`; `mod_mesh_read_gmsh!`; `exact_geometry.jl` (called at `mesh.jl`); deck `:exact_geometry`.

Step 2 — basis, quadrature and metric terms

On the reference square the basis is the tensor product of the degree-N Lagrange polynomials through the LGL nodes, $\psi_{ij}(\xi, \eta) = \ell_i(\xi)\ell_j(\eta)$, and the displacement is

$$\boldsymbol{u}_h(\boldsymbol{x}) = \sum_a \psi_a(\boldsymbol{x})\, \boldsymbol{u}_a, \qquad a = \mathrm{conn}(e, i, j). \tag{18}$$

Integrals use the LGL rule on the same nodes (collocation, $Q = N$; Jexpresso's `Inexact` quadrature), weights $\omega_k\omega_l$. The map $\boldsymbol{x}(\xi, \eta) = \sum \psi_{ij}\boldsymbol{x}_{ij}$ gives the metric terms $\partial\xi/\partial x, \ldots$ and J at every node; with them

$$\frac{\partial \psi_{ij}}{\partial x}\Big|_{kl} = \ell_i'(\xi_k)\ell_j(\xi_l)\,\xi_x + \ell_i(\xi_k)\ell_j'(\xi_l)\,\eta_x, \qquad \int_{\Omega_e} g\,\mathrm{d}\Omega \approx \sum_{k,l} \omega_k\omega_l\, J_{kl}\, g_{kl}. \tag{19}$$

On a boundary edge, the surface Jacobian $J^f = |\mathrm{d}\boldsymbol{x}/\mathrm{d}\xi|$ and the outward normal come from the tangent of the same degree-N interpolant, so they are right on the curved arcs.

Code: `sem_setup`; `build_metric_terms!` (2D, with the boundary normals and J^f at `metric_terms.jl`).

Step 3 — element matrices and direct stiffness summation

Inserting $\boldsymbol{u}_h$ and $\boldsymbol{v} = \psi_a \boldsymbol{e}_\alpha$ into (7) with the Voigt form $\boldsymbol{\sigma} = \mathbf{D}\,\boldsymbol{\varepsilon}$ ($\boldsymbol{\varepsilon} = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})$, $\gamma_{xy} = 2\varepsilon_{xy}$, $\mathbf{B}_a = [\partial_x\psi_a\ 0;\ 0\ \partial_y\psi_a;\ \partial_y\psi_a\ \partial_x\psi_a]$) gives the stiffness $\mathbf{K}_{ab} = \sum_e \int_{\Omega_e} \mathbf{B}_a^{\mathsf{T}}\mathbf{D}\,\mathbf{B}_b\,\mathrm{d}\Omega$. Jexpresso builds it from the directional element matrices

$$L_e^{\alpha\beta}[a, b] = \int_{\Omega_e} \partial_\alpha\psi_a\, \partial_\beta\psi_b\,\mathrm{d}\Omega \approx \sum_{k,l} \omega_k\omega_l J_{kl}\, \partial_\alpha\psi_a \partial_\beta\psi_b\big|_{kl}, \tag{20}$$

the components of Jexpresso's Laplacian element matrix ($L_e^{xx} + L_e^{yy}$ is exactly it), assembled by direct stiffness summation (DSS) into sparse global matrices $S_{\alpha\beta} = \mathrm{DSS}(L^{\alpha\beta})$. For the isotropic material, $\mathbf{D} = K\mathbf{D}_{\mathrm{vol}} + \tilde{G}\mathbf{D}_{\mathrm{dev}}$ with $\mathbf{D}_{\mathrm{vol}} = \left[\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right]$, $\mathbf{D}_{\mathrm{dev}} = \left[\begin{smallmatrix} 4/3 & -2/3 & 0 \\ -2/3 & 4/3 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$, so in the flat layout $\mathbf{u} = [\mathbf{u}_x; \mathbf{u}_y]$

$$\mathbf{K}(\tilde{G}) = K\,\mathbf{A}_{\mathrm{vol}} + \tilde{G}\,\mathbf{A}_{\mathrm{dev}}, \quad \mathbf{A}_{\mathrm{vol}} = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{yx} & S_{yy} \end{bmatrix}, \quad \mathbf{A}_{\mathrm{dev}} = \begin{bmatrix} \frac{4}{3}S_{xx} + S_{yy} & S_{yx} - \frac{2}{3}S_{xy} \\ S_{xy} - \frac{2}{3}S_{yx} & S_{xx} + \frac{4}{3}S_{yy} \end{bmatrix}, \tag{21}$$

$S_{yx} = S_{xy}^{\mathsf{T}}$, assembled once. When the tangent varies from point to point (plasticity) the four displacement blocks are built with $\mathbf{D}_{kl}$ inside the quadrature sum and assembled the same way. With LGL collocation the quadrature points are the nodes, so $\nabla\psi_a$ vanishes at the point (k, l) unless node a shares its row or its column: $2N + 1$ of the $(N + 1)^2$ basis functions. Both element routines loop over those pairs only — $(2N + 1)^2 = 81$ products per point at $N = 4$ instead of $(N + 1)^4 = 625$ — accumulating each element in a dense buffer that is scattered once into the element-matrix layout `DSS_laplace_sparse` reads.

Code: `build_laplace_matrix` and its components `build_gradient_product_matrices`; variable tangent `build_tangent_stiffness_matrices`; the non-zero gradients `_element_basis_gradients!`; DSS `DSS_laplace_sparse`; assembly of (21) `solid_sem_operators`, `solid_tangent_matrix`.

Step 4 — the load vector

The pressure (3) is integrated on every boundary edge of Γ_a with the LGL rule of the edge and the curved-edge Jacobian, then summed onto the nodes:

$$\mathbf{f}_a = \sum_{\text{edges}} \sum_{k} \omega_k \, J_k^f \, (-p\,\boldsymbol{n}_k)\, \delta_{a,\text{node}(k)}. \tag{22}$$

A body force would enter through the lumped (diagonal) LGL mass matrix, $\mathbf{f}_a \mathrel{+}= M_a \boldsymbol{b}(\boldsymbol{x}_a)$.

Code: `solid_boundary_conditions!` with Jexpresso's `compute_segment_integral!` and `DSS_segment_integral!`; body force `solid_add_body_force!`.

Step 5 — the symmetry conditions

Each roller fixes one Cartesian component ($u_y = 0$ on Γ_x, $u_x = 0$ on Γ_y); together they remove both translations and the rotation. The constrained DOFs c are eliminated and the solve is on the free block f, which stays symmetric positive definite:

$$\mathbf{K}_{ff}\,\mathbf{u}_f = \mathbf{f}_f - \mathbf{K}_{fc}\,\bar{\mathbf{u}}_c, \qquad \mathbf{u}_c = \bar{\mathbf{u}}_c. \tag{23}$$

$\mathbf{K}_{ff}$ depends on Δt only through $\tilde{G}$, so one sparse Cholesky factorisation serves every step (a second one the instantaneous step).

Step 6 — marching in time: the history as a nodal field

The discrete strain is linear in the nodal displacements and (10) is linear and starts from zero, so at every quadrature point $\boldsymbol{q}_m = \operatorname{dev}\boldsymbol{\varepsilon}(\mathbf{w}_m)$ of a *nodal* field $\mathbf{w}_m$ obeying the same recursion. With $\mathbf{v} \cdot \mathbf{A}_{\text{dev}}\mathbf{z} = \int \boldsymbol{\varepsilon}(\mathbf{v}) : 2\operatorname{dev}\boldsymbol{\varepsilon}(\mathbf{z})$, (11) becomes, for every step $n \to n + 1$,

$$\begin{aligned} (K\mathbf{A}_{\text{vol}} + \tilde{G}\mathbf{A}_{\text{dev}})\mathbf{u}^{n+1} &= \mathbf{f}(t_{n+1}) - G\,\mathbf{A}_{\text{dev}} \sum_m \mu_m (d_m \mathbf{w}_m^n - \gamma_m \mathbf{u}^n), \\ \mathbf{w}_m^{n+1} &= d_m \mathbf{w}_m^n + \gamma_m (\mathbf{u}^{n+1} - \mathbf{u}^n), \end{aligned} \tag{24}$$

which reproduces the quadrature-point recursion to round-off.

Code: the march `nodal path`; driver `quasistatic_solid_solve!`.

Algorithm 1 One run of the viscoelastic cylinder (nodal history)

assemble $\mathbf{A}_{\mathrm{vol}}, \mathbf{A}_{\mathrm{dev}}$ (Step 3); $\mathbf{u} \leftarrow \mathbf{0}$, $\mathbf{w}_m \leftarrow \mathbf{0}$
if loads on at t_0: one step of (24) with $\Delta t = 0$ ($d = \gamma = 1$, $\tilde{G} = G$) ▷ glassy response
for $n = 0, 1, \ldots$ **do**
    $d_m, \gamma_m \leftarrow$ (10); $\tilde{G} \leftarrow G(\mu_0 + \sum \mu_m \gamma_m)$
    $\mathbf{f} \leftarrow$ pressure load (Step 4) $- G\mathbf{A}_{\mathrm{dev}} \sum_m \mu_m (d_m \mathbf{w}_m - \gamma_m \mathbf{u})$
    solve the free block (Step 5) with the cached factorisation of $\mathbf{K}(\tilde{G})$
    $\mathbf{w}_m \leftarrow d_m \mathbf{w}_m + \gamma_m(\mathbf{u}^{n+1} - \mathbf{u}^n)$; $\mathbf{u} \leftarrow \mathbf{u}^{n+1}$
    post-process (Step 8)
end for

Algorithm 2 One increment on the quadrature-point path

$\mathbf{u}_c \leftarrow \bar{\mathbf{u}}_c(t_{n+1})$ ▷ prescribed displacements
repeat
    $\boldsymbol{\varepsilon}_{kl} \leftarrow$ element gradient of $\mathbf{u}$ at every LGL point ▷ `_∇f!`
    $(\boldsymbol{\sigma}_{kl}, \mathbf{D}_{kl}, H^{\mathrm{tr}}_{kl}) \leftarrow$ material point$(\boldsymbol{\varepsilon}_{kl}, H^n_{kl}, \Delta t)$
    $\mathbf{f}^{\mathrm{int}}_a = \sum_e \sum_{kl} \omega_k \omega_l J_{kl} \nabla\psi_a \cdot \boldsymbol{\sigma}_{kl}$ ▷ weak divergence + DSS
    $\mathbf{r}_f \leftarrow \mathbf{f}^{\mathrm{ext}}_f - \mathbf{f}^{\mathrm{int}}_f$
    $\mathbf{K}_T \leftarrow \sum_e \int \mathbf{B}^{\mathsf{T}} \mathbf{D}_{kl} \mathbf{B}$; $\mathbf{u}_f \leftarrow \mathbf{u}_f + \mathbf{K}^{-1}_{T,ff} \mathbf{r}_f$
until $\|\mathbf{r}_f\| \le 10^{-10} \max(\|\mathbf{f}^{\mathrm{ext}}\|, \|\mathbf{f}^{\mathrm{int}}\|)$
$H^{n+1} \leftarrow H^{\mathrm{tr}}$ ▷ accept the history

Step 7 — the general path: quadrature-point history and Newton

For a nonlinear material (or with `:solid_history => "quadrature"`) the history H lives at every quadrature point and each increment solves $\mathbf{R}(\mathbf{u}) = \mathbf{f}^{\mathrm{int}}(\mathbf{u}; H^n) - \mathbf{f}^{\mathrm{ext}} = \mathbf{0}$ by Newton's method with the consistent tangent: An increment that has not converged after `:newton_maxit` iterations is halved and the two halves are solved in turn, recursively, up to `:newton_max_cuts` times; `solid_history.csv` logs the iterations and cuts of every step. With the viscoelastic model this path reproduces (24) to round-off; with J_2 plasticity it takes about four iterations per increment on the cylinder and about six on the perforated plate (§7).

Code: `quadrature path`; Newton `newton!` and the halving `advance!`; strain `solid_qp_strains!` (with `_∇f!`); internal force `solid_internal_force!` (the contraction of `_expansion_visc!`, then `DSS_rhs!`).

Step 8 — post-processing and output

Nodal stresses (and history fields) are the mass-weighted averages of the element values at shared nodes: element gradients by `_∇f!`, summed by `DSS_rhs!`, divided by the lumped mass M_a — the SEM nodal projection. Errors are measured in the same LGL norm, $\|\boldsymbol{e}\|^2 = \sum_a M_a |\boldsymbol{e}_a|^2$. Fields go to VTK through Jexpresso's `write_output` (with the displacement also as a vector, `:vtk_vectors`, for ParaView's *Warp By Vector*); `solid_history.csv` logs every step; `:lplot_deformed` draws the deformed grid.

Code: `solid_nodal_gradient!`, `solid_qp_to_nodes!`; `write_vtk` (its `:vtk_vectors` option); `plot_deformed_mesh`.

5 Initialisation

The run starts from the undeformed, unstressed state with no history: $\mathbf{u} = \mathbf{0}$, $\mathbf{w}_m = \mathbf{0}$ (or $H = 0$: $\boldsymbol{q}_m = \boldsymbol{e}^n = \mathbf{0}$, $\boldsymbol{e}^p = \mathbf{0}$, $\bar{e}^p = 0$). `initialize` declares the state fields (`ux, uy, sxx, syy, sxy, szz`, plus any history fields to plot) and zeroes them; the solver refuses a non-zero initial displacement, because the history that produced it would not be known.

Two load conventions, chosen by `:linitial_elastic_solve`:

- **true** (default): the pressure is on at $t = 0^+$. One step with $\Delta t = 0$ ($d_m = \gamma_m = 1$, $\tilde{G} = G\sum\mu$) gives the instantaneous, glassy response, $u_A(0^+) = 0.0381333$ (Lamé); the creep follows. This is the problem the closed form solves.
- **false**: start unloaded; the full pressure acts from the first increment. Through (10) that is a load ramped over $[0, \Delta t]$ — what FEAP does when a load is switched on at step 1, and why the paper's Fig. 5.2 reads $u_A = 0.058$ at $t = 1$ where the closed form gives 0.073.

Code: `initialize`; the zero-state check and the instantaneous step `quasistatic_solid.jl`; the switch `:linitial_elastic_solve`.

6 Results and verification

6.1 Viscoelastic cylinder ($N = 4$, 4×8 elements)

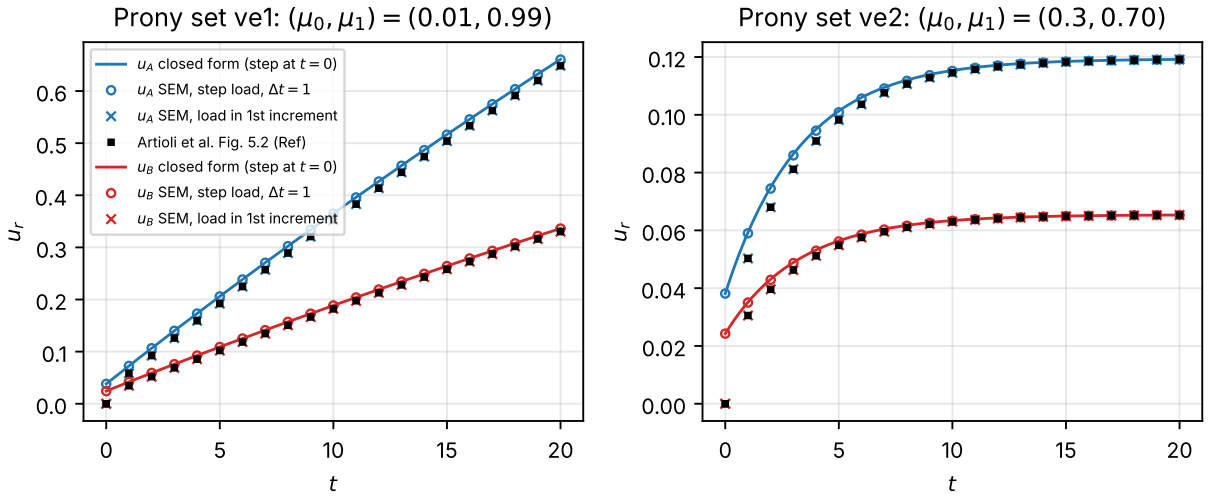


Figure 2: Radial displacement at A and B. Lines: closed form (Heaviside load). Circles: SEM with the load on at $t = 0^+$ and $\Delta t = 1$ — on the closed form within the second-order time error. Crosses: SEM with the load applied in the first increment (FEAP's convention) — on the paper's Fig. 5.2 points (squares, read from the vector graphics of the arXiv PDF).

6.2 Elastic–perfectly plastic cylinder (the history at the quadrature points)

The same quarter cylinder, J_2 with the material of Artioli et al. §4.3 ($E = 7000$, $\nu = 0.3$, $\sigma_{y0} = 24.3$, no hardening), $N = 4$, 8×8 elements, its bore displaced radially to $\delta = 0.05$ in 50 increments. Up to first yield the solution is Lamé's ($p_e = 10.5047$); with the whole wall flowing, the pressure levels off at the exact limit $p_L = \frac{2}{\sqrt{3}}\sigma_{y0} \ln \frac{b}{a} = 19.4492$ (Hill 1950).

check	result	reference
$t = 0^+$, u_A	0.0381333412	Lamé 0.0381333333
spatial error alone, $N = 2, 3, 4, 5, 6$	2.3×10^{-3}, 2.5×10^{-6}, 6.0×10^{-7}, 4.1×10^{-9}, 1.0×10^{-9}	time-discrete reference
time error, $\Delta t = 1 \to \frac{1}{8}$, exponential (ve1/ve2)	rates 1.98–2.04 / 1.97–2.00	closed form
time error, midpoint (ve1) / backward Euler (ve2)	rates 2.00–2.01 / 0.91–0.98	closed form
quadrature-point path vs nodal path	7×10^{-13} (relative)	round-off
paper Fig. 5.2, $\max\lvert u - u_{\text{paper}}\rvert$ (ve1 / ve2)	1.1×10^{-4} / 2.4×10^{-5}	figure resolution

Table 2: Verification of the viscoelastic case (`test/test_viscoelastic_cylinder.jl`, 23/23). Errors are relative to $\max u_A$ unless stated.

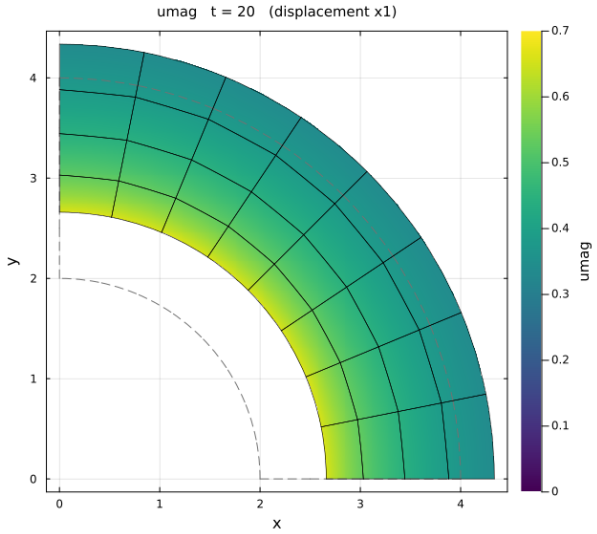


Figure 3: The deformed grid at $t = 20$ (ve1), true scale, coloured by $|\boldsymbol{u}|$, over the undeformed outline (dashed): the bore has crept from $r = 2$ to 2.66. Drawn by `plot_deformed_mesh` (`:lplot_deformed`).

6.3 Seeing the solid deform

Both cases write the displacement also as a vector `u` (`:vtk_vectors`). In ParaView: open `simulation.pvd`, apply *Warp By Vector* with vector `u` (scale 1 for the viscoelastic cylinder, ∼10 for the plastic one), colour by any field (`szz`, `svm`, `ep_bar`, ...) and press *Play*. Without ParaView, `:lplot_deformed => true` writes `deformed-<field>-it<k>.png` and `deformed.gif` (Figs. 3, 4).

Code: `test/test_viscoelastic_cylinder.jl`, `test/test_plastic_cylinder.jl`.

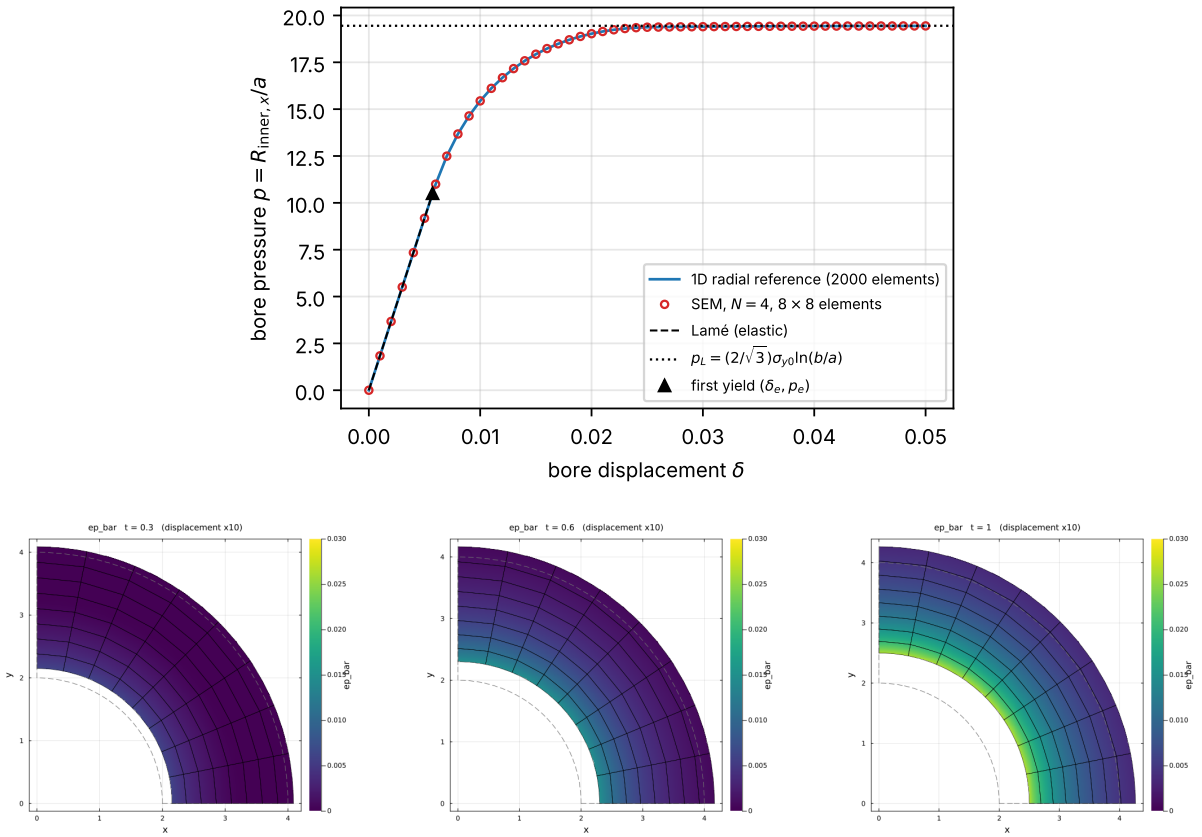


Figure 4: Top: bore pressure against bore displacement — SEM, the 1D radial reference (same constitutive routine and increments), Lamé and the exact limit. Bottom: the deformed grid ($\times 10$) coloured by the accumulated plastic strain $\bar{e}^p$, a quadrature-point history value projected onto the nodes, at $t = 0.3$, 0.6 and 1: the plastic zone spreads from the bore until the whole wall has yielded.

check	result
J_2 point, cyclic uniaxial strain, isotropic + kinematic hardening, vs exact	1.3×10^{-16}
consistent tangent vs finite differences	$\leq 1.7\times10^{-11}$
bore pressure before first yield vs Lamé	1.9×10^{-7}, $\bar{e}^p = 0$
bore pressure vs 1D radial reference, all increments	$1.3\times10^{-4}\,p_L$
$p(\delta = 0.05)$	19.44145 (reference 19.44143, $p_L = 19.44917$)
Newton iterations per increment	3.8 on average, at most 5

Table 3: Verification of the plasticity case (`test/test_plastic_cylinder.jl`, 16/16).

7 The perforated plastic plate (§4.3 of the paper)

The paper's second inelastic benchmark puts the quadrature-point path of Step 7 through a genuinely two-dimensional plastic flow. A strip $2L \times 2H = 200 \times 360$ mm with a central hole of diameter $2R = 100$ mm, in plane strain, made of the elastic–perfectly plastic J_2 material of §3 with $E = 7000$ kg/mm^2, $\nu = 0.3$, $\sigma_{y0} = 24.3$ kg/mm^2 and no hardening, is stretched along its length: the normal displacement is prescribed on the symmetry lines, on the lateral edges and on the ends, which are pulled apart by $\delta = 2$ mm in 400 equal increments. A quarter is modelled (Fig. 5).

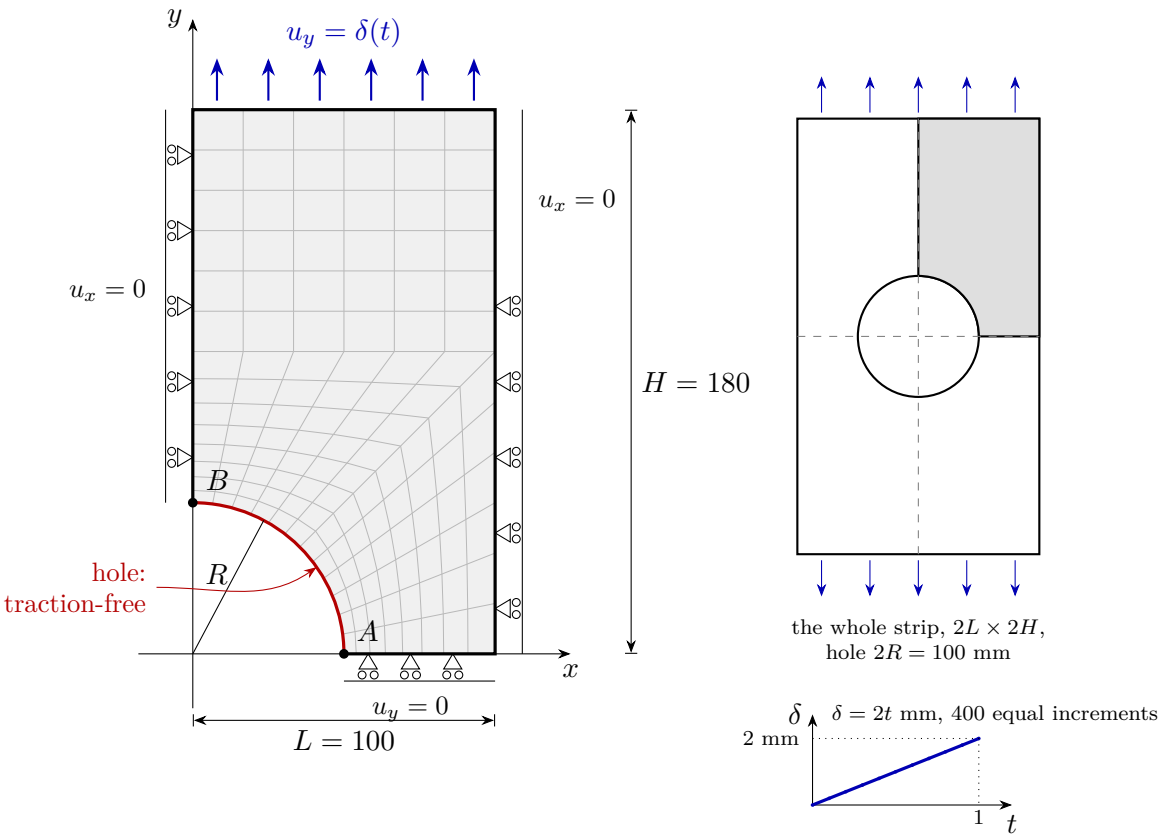


Figure 5: The perforated plate. Left: the modelled quarter, plane strain, with its spectral-element grid — 132 quadrilaterals in three structured blocks graded towards the hole, the hole curved onto its circle — and the boundary conditions: the top edge pulled up by $\delta(t)$ (tangentially free), the lateral edge and the two symmetry lines on rollers, the hole traction-free. $A = (R, 0)$ and $B = (0, R)$ are the points of Table 1 of the paper. Right: the whole strip, the quarter shaded; the imposed displacement, $\delta = 2t$ mm in 400 equal increments.

Code: the case `problems/Elasticity/perforated_plate2d`: deck `user_inputs`, material and geometry `perforated_plate_properties`, boundary conditions `user_solid_bc`, the paper's numbers `perforated_plate_paper_reference`; grid `structured_perforated_plate` (three transfinite blocks graded towards the hole, the hole curved back onto its circle by `:exact_geometry`).

Discretisation and solution. Nothing is specific to the plate: the grid, basis and metric terms of Steps 1–2, the tangent of Step 3 rebuilt and refactorised at every Newton iteration, the prescribed top displacement as a time-dependent Dirichlet value of Step 5, the history $\{\boldsymbol{e}^p, \bar{e}^p\}$ at each of the

tag	where	condition (`user_solid_bc`)
`top`	$y = H$	$u_y = \delta(t) = 2t$ mm, u_x free (the load)
`right`	$x = L$	$u_x = 0$, u_y free (the lateral restraint)
`left`	$x = 0$, $y \geq R$	$u_x = 0$ (symmetry)
`bottom`	$y = 0$, $x \geq R$	$u_y = 0$ (symmetry)
`hole`	$r = R$	traction-free
grid: 132 quadrilaterals, $N = 4$: 2217 nodes, 4434 DOFs (164 constrained); $t \in [0, 1]$, $\Delta t = 1/400$		

Table 4: Boundary conditions and discretisation of the perforated plate. The material is rate independent, so t is a load parameter. The corners join a u_x and a u_y condition, so every constrained component belongs to one tag and the reaction on `top` is the force the top edge carries.

132×25 quadrature points, and Algorithm 2 at every increment. The reactions are the residual on the constrained components, $\mathbf{R}_c = \mathbf{f}_c^{\text{int}} - \mathbf{f}_c^{\text{ext}}$, written as the nodal fields `rx`, `ry` (zero elsewhere) and summed per tag into `solid_history.csv`. Starting from the unloaded state (`:linitial_elastic_solve => false`) the whole run takes about three minutes, VTK output every four increments included, on one core.

Code: reaction fields `rx, ry`; per tag `reactions`.

Results. Table 5 compares the displacements of A and B at the end of the loading with the paper's Table 1: Jexpresso's order-4 elements land on the paper's reference solution (an overkill mixed u–p–ε^v finite-element computation) to 0.07 % and 0.27 %, closer than any of the paper's quadratic VEM or FEM. The reaction on the top edge (Fig. 6) rises elastically, bends as the plastic zone spreads from the hole (Fig. 7) across the ligament to the lateral edge, and levels off at the limit load. Over the elastic steps the paper's reaction sums are exactly ten times Jexpresso's (9.990–10.010 across its five methods) while the displacements agree: the paper's force is that of a plate 10 mm thick, a thickness it does not state, and Jexpresso's force per unit thickness is multiplied by 10 to be drawn on the paper's axes. At the end of the loading it is 19 507 kg against the paper's 19 557–19 755 kg, 0.25–1.25 % below its quadratic methods — displacement-based elements approach a limit load from above — and nearly flat: it rises by 0.05 % from step 200 to step 400, the paper's curves by 0.16–0.70 %.

	u_A [mm] (Table 1)	v_B [mm] (Table 1)	Newton iterations per increment (Table 2)
Jexpresso SEM, $N = 4$, 132 elements	**2.7390**	**1.8539**	**5.76**
paper, reference (mixed FEM, overkill)	2.741	1.859	—
paper, VEM $k = 2$, Quad	2.720	1.851	6.43
paper, VEM $k = 2$, Tri	2.719	1.852	6.11
paper, VEM $k = 2$, Voronoi	2.714	1.852	6.21
paper, FEM Q9	2.733	1.853	6.70
paper, FEM T6	2.719	1.853	6.11

Table 5: The perforated plate at the end of the loading ($\delta = 2$ mm): horizontal displacement of $A = (R, 0)$, vertical displacement of $B = (0, R)$, and the average number of Newton iterations per load increment (Jexpresso: tolerance $\|\mathbf{r}\| \leq 10^{-10}\|\mathbf{f}\|$, 9 at most, no increment cut).

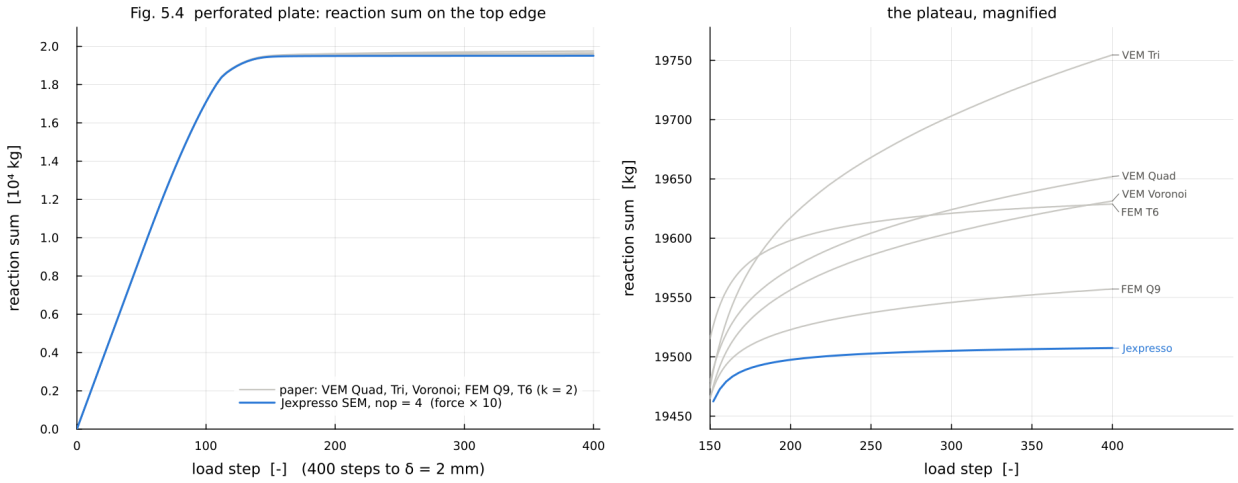


Figure 6: The paper's Fig. 5.4 redrawn by `tools/artioli2017_figures.jl`: the reaction sum on the top edge against the load step. Grey: the paper's five quadratic methods, read from the vector graphics of the arXiv PDF. Blue: Jexpresso, summed from the nodal reaction field `ry` of the VTK files, ×10 (the paper's plate is 10 mm thick). Right: the plateau, magnified.

Redrawing the paper's figures. `tools/artioli2017_figures.jl` runs the three computations at the paper's settings (the viscoelastic cylinder with both Prony sets and FEAP's load convention, and the plate), reads every `iter_<k>.pvtu` listed in `simulation.pvd` with `tools/vtu_reader.jl` — points, fields and cells — and writes Fig. 5.2(a,b), Fig. 5.4, Table 1 with Table 2, the plastic zone, and an animation of the plate deforming beside its reaction curve:

```
julia --project=. tools/artioli2017_figures.jl            # run, then plot
julia --project=. tools/artioli2017_figures.jl --no-run   # replot earlier runs
```

Code: `figure_5_4`, the force scale `PAPER_FORCE_SCALE`; the reader `read_vtu`; the test `test/test_perforated_plate.jl`.

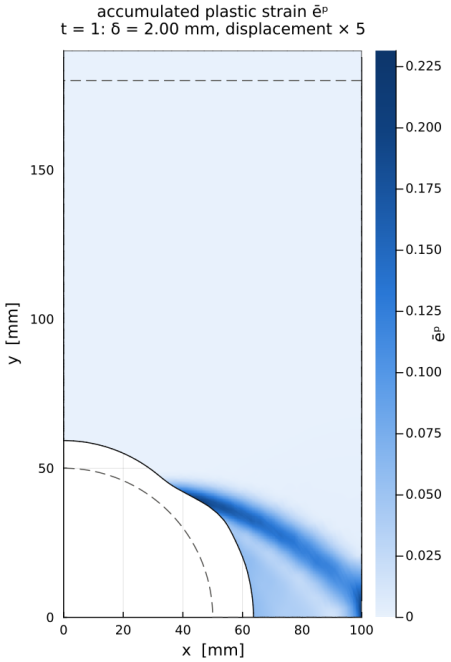


Figure 7: The quarter plate at the end of the loading, displacement $\times 5$, coloured by the accumulated plastic strain $\bar{e}^p$ (a quadrature-point history value projected onto the nodes), over the undeformed outline (dashed): the ligament between the hole and the lateral edge has yielded through, and A has moved towards the lateral edge as the ligament thins.

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